

I need another small test case for Additive Phylogeny problem

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Allison Cooper · a day ago

My program works for small test cases I make up myself, but I can't get it to work for the extra data set or the assignment. My latest test case is for a tree with 3 internal nodes and 6 leaves. 3 leaves share a parent, 2 other leaves share a parent, and one leaf has no sibling leaves. Any suggestions on what kind of tree I should use in order to trip up my program to let me know what I'm doing wrong?

Thank you!!

0 · flag



John D. Cox · a day ago

6					
0	3	8	15	25	24
3	0	9	16	26	25
8	9	0	15	25	24
15	16	15	0	24	23
25	26	25	24	0	17
24	25	24	23	17	0

0->6:1

1->6:2

2->7:4

3->8:7

4->9:9

5->9:8

6->0:1

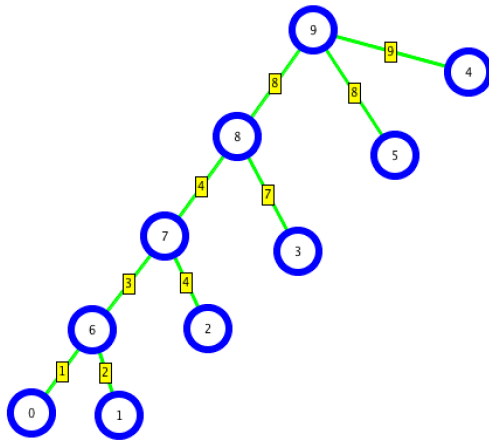
6->1:2

6->7:3

7->2:4

7->6:3

7->8:4
 8->3:7
 8->7:4
 8->9:8
 9->4:9
 9->5:8
 9->8:8



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William J. F. McLean · a day ago

While it is certainly desirable that your algorithm should be able to cope with test cases of arbitrary structure, in practice the examples provided all appear to be limited to nodes of degree 3 (internal nodes) and of degree 1 (leaf nodes). You definitely shouldn't have a node of degree 2 (the one connected to the leaf with no siblings in your description), because in that case the edges could simply be combined and the node eliminated.

This problem has been discussed in previous threads - perhaps you'll find some helpful suggestions there:

https://class.coursera.org/bioinformatics2-001/forum/thread?thread_id=99

https://class.coursera.org/bioinformatics2-001/forum/thread?thread_id=110

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Allison Cooper · a day ago

@John, the good news is ... I passed your test. The bad news is I still don't know my problem. Thanks for trying!

@William, I hadn't seen that second thread. I'll take a look. I think I poorly described the sample I created myself. Here it is:

```

6
0  5  6  18  15  8
5  0  7  19  16  9
6  7  0  20  17  10
18 19 20  0  7  12
15 16 17  7  0  9
8  9 10 12  9  0

```

```

0->6:2
1->6:3
2->6:4
3->7:5
4->7:2
5->8:1
6->1:3
6->0:2
6->2:4
6->8:5
7->3:5
7->4:2
7->8:6
8->7:6
8->6:5
8->5:1

```

Here's the beginning of my output for the extra data set: My first line matches the Output provided to us (different created node) but that is all:

```

0->29:745
1->33:1127
2->32:788
3->30:409
4->31:280
5->35:2455
6->33:492
7->34:657
8->35:957
9->36:2049
10->37:280

```

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William J. F. McLean · a day ago 🔒

I notice that your output is correct, in terms of limb length, for nodes 0, 2, 3, 4, 6, 7 & 10 so clearly things are working correctly some of the time. Given that nodes 0 - 28 are leaves, you should be able to get the correct edge length if your Limb(D, n) function is operating as it should. That is definitely where you should begin looking for a bug.

Edit:

My reply was a bit hasty - on reconsideration, your Limb() function is probably ok.

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John D. Cox · a day ago 🔒

The length 1127 should be 156 but you have added 3 paths into this length (156 + 87 + 884). Not sure how or if this will help you debug :)

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Allison Cooper · a day ago 🔒

Thank you both. I will look again tomorrow.

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Prabhjot Mundi Signature Track · a day ago 🔒

I've really struggled with this problem. I just don't understand exactly what is being done in the recursive algorithm being described. Could someone explain to me how for a larger distance matrix than 4x4, the tree is put together? If the matrix is 20x20, is the "bare matrix" that they talk about 19x19 in each step, or does it progressively keep getting smaller as you prune off leaves? Are the i and k leaves that they talk about finding (for which the newnode is going to be along at distance of Dbare[i,j] always going to be 1 of 19 possibilities with each step, or does the matrix keep getting smaller such that you would be considering fewer and fewer i's and k's?

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Prabhjot Mundi Signature Track · a day ago 🔒

I have little background in computer science/algorithms, so the more dumbed down the explanation, the better!

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Allison Cooper · 11 hours ago 🔒

Whew! I finally figured out my problem. I had a faulty path finder function. What worked for me as a test case was to use the 8x8 upper left corner of the input matrix D.

@Prabhjot, here is my pseudocode. Note: n is fixed from input; m varies and corresponds to how much of the matrix D you are looking at: first 2x2, then 3x3, etc. To start the whole process, though, you run the program with m=n.

```
additivePhylogeny(D,n,m)
```

```
if m = 2:
```

```
    make 0 and 1 connect with distance = D[0,1]
```

```
    return
```

```
let limbLen, i, and k be the values found in limbLength(D,m,m-1) # node to attach: m-1
```

```
# Note: only use mxm matrix to find limb length of m-1
```

```
distToAttach = D[i,m-1] - limbLen # this is the distance from i to where m-1 should attach
```

```
additivePhylogeny(D,n,m-1) # recursive call creates the tree with one fewer node
```

```
# Note: the remaining steps attach m-1 to the tree created in the recursive call
```

```
let path be the path from i to k (from above) INCLUDING distance of each step
```

```
go along path, keeping track of total distance, until reaching distToAttach
```

```
if you land on a node, connect m-1 to this node with its limb length from above
```

```
if you land between nodes, create a new node between them and attach m-1 there
```

```
# Note: if you create a new node, you'll have to make several updates to the tree
```

I suggest you draw out one of the examples above or use the 8x8 part of D from the extra data set.

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